CALCULUS I - AY2024-25

Chapter 3: Applications of Derivatives

Summary

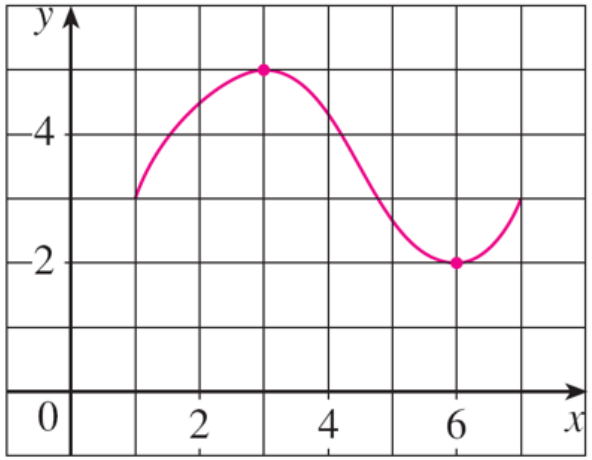
One of the most important applications of the derivative is its use as a tool for finding the optimal (best) solutions to problems.
We study the following topics.

- Maximum and Minimum (Extreme) Values of Functions
- The Mean Value Theorem
- Derivative Tests
- Curve Sketching
- Optimization Problems
- Newton's Method
- Antiderivatives

Outline

1. Maximum and Minimum Values
2. The Mean Value Theorem
3. Derivative Tests
4. **Curve Sketching: a self-study**
5. Applied Optimization Problems
6. L'Hospital's Rule
7. Newton's Method
8. Antiderivatives

Let us consider the figure below.



We see that the highest point on the graph of the function $f(x)$ is the point $(3, 5)$, while the lowest point is the point $(6, 2)$. We say that $f(3) = 5$ is the absolute maximum of $f(x)$ and $f(6) = 2$ is the absolute minimum.

In general, we use the following definition.

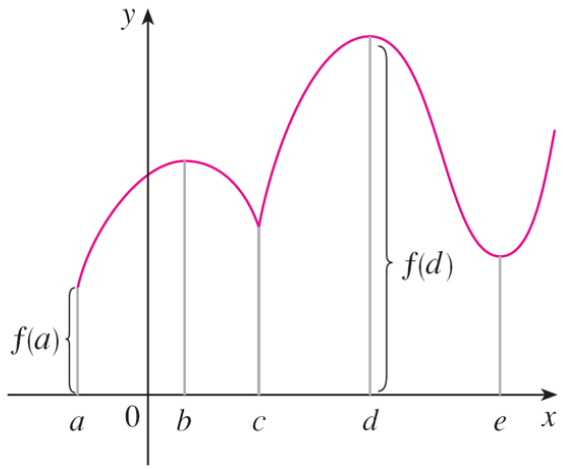
Definition. Let c be a number in the domain D of a function $f(x)$. Then $f(c)$ is the

- absolute maximum value of $f(x)$ on D, if $f(c) \geq f(x)$ for all $x \in D$
- absolute minimum value of $f(x)$ on D, if $f(c) \leq f(x)$ for all $x \in D$.

- An absolute maximum or minimum is sometimes called a global maximum or minimum.

- The maximum and minimum values of $f(x)$ are called extreme values of $f(x)$.

The figure below shows the graph of a function $f(x)$ with absolute max at $x = d$ and absolute min at $x = a$.



If we consider values of x near b (for instance, if we restrict our attention to the interval (a, c)), then $f(b)$ is the largest values in the interval and is called a local maximum value of $f(x)$.

Likewise, $f(c)$ is called a local minimum value of $f(x)$. The function $f(x)$ also has a local min at $x = e$.

In general, we have the following definition.

Definition. The number $f(c)$ is a

- local maximum value of $f(x)$ if $f(c) \geq f(x)$ when x is near c
- local minimum value of $f(x)$ if $f(c) \leq f(x)$ when x is near c.

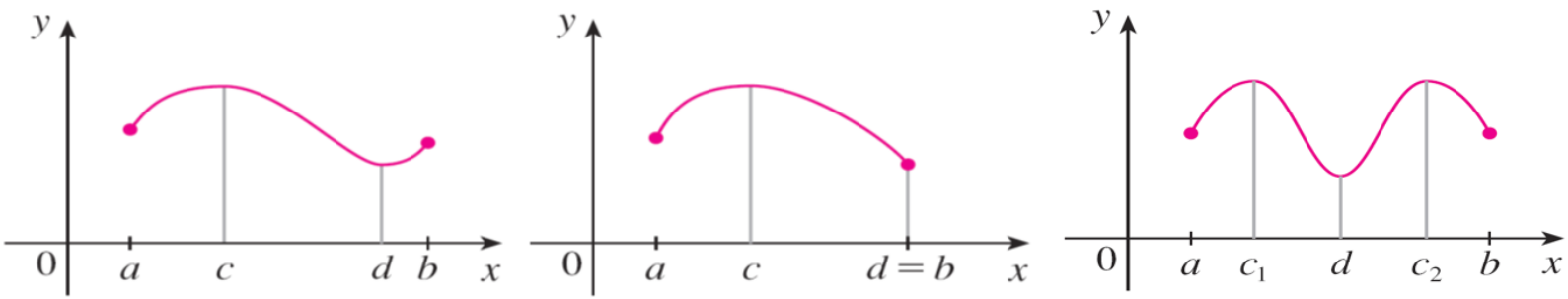
To be more specific, if we say that something is true near c, we mean that it is true on some open interval containing c.

The following result gives conditions under which a function is guaranteed to possess extreme values.

Theorem (The Extreme Value Theorem)

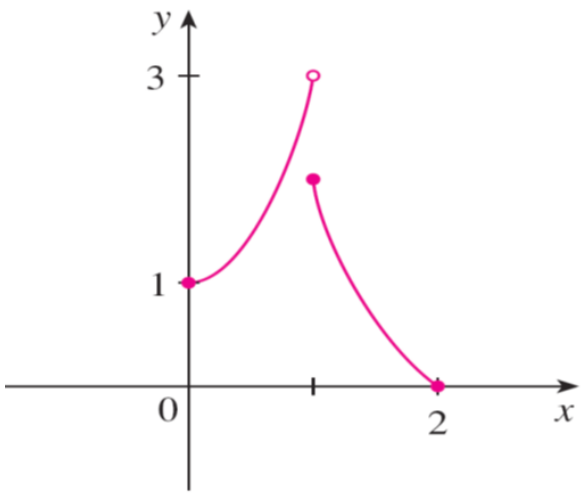
If $f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ attains both an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some number c and d in $[a, b]$.

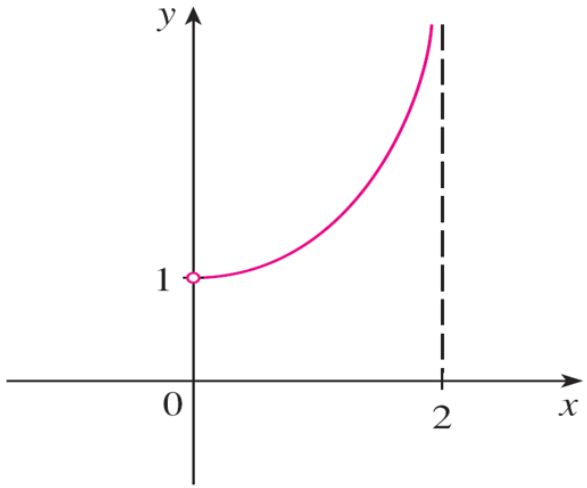
The Extreme Value Theorem is illustrated in the figures below,



Again, a function can attain extreme values more than once.

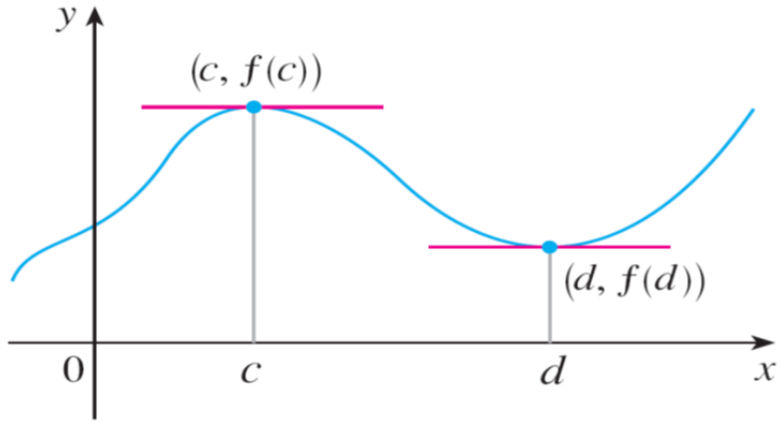
Note. Conditions continuity and closed interval in the Extreme Value Theorem are essential; that is if either of these conditions is violated, then a function may not have extreme values.





- Left: A function is not continuous on the interval. It has no max value. The range of $f(x)$ is $[0, 3)$, it can never attains the value 3.
- Right: A function is continuous on the open interval $(0, 2)$. It does not have neither a max nor a min value. The range of $f(x)$ is $(1, \infty)$.

The Extreme Value Theorem does not tell us how to find these extreme values. However, we can start by looking for local extreme values.



Theorem (Fermat's Theorem)

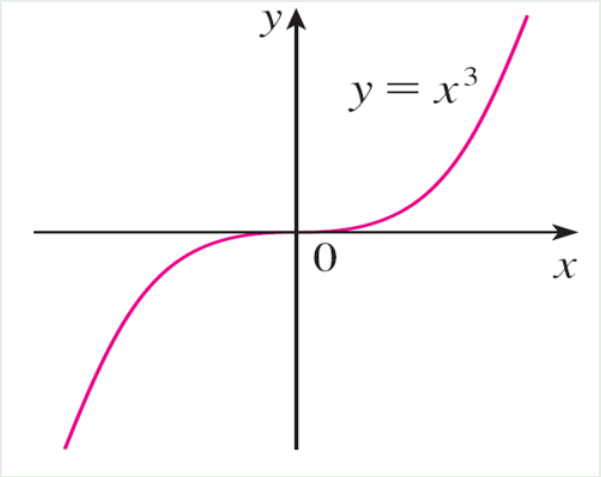
If f has a local maximum or minimum at an interior point c of its domain, and if $f'(c)$ exists, then

$$f'(c) = 0.$$

The converse of Fermat's Theorem, in general, is not true.

Example

Consider $f(x) = x^3$ at $x = 0$.



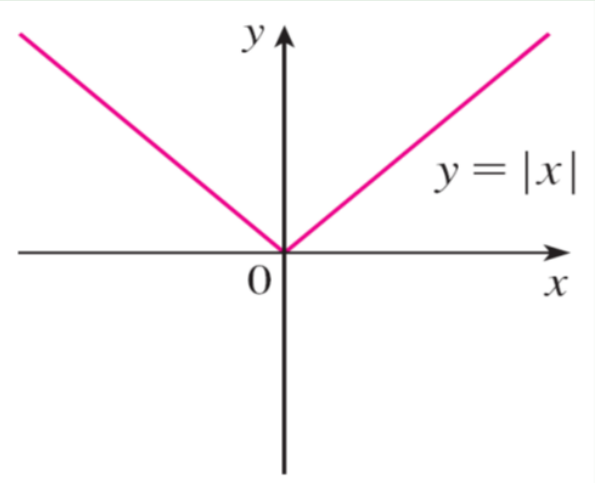
We have $f'(x) = 3x^2$ and $f'(0) = 0$, but $f(x)$ has no maximum or minimum at 0.

The condition $f'(0) = 0$ simply means that the curve $y = x^3$ has a horizontal tangent at $(0, 0)$.

We look at another example.

Example

The function $f(x) = |x|$



It has (local and absolute) minimum value at $x = 0$, but $f'(0)$ does not even exist.

The above examples show that we must be careful when using Fermat's Theorem.

To characterize the points above, we need the following definition.

Definition. An interior point c of the domain of a function f is called a critical point, if either $f'(c) = 0$ or $f'(c)$ does not exist.

In terms of critical numbers, Fermat's Theorem can be rephrased as follows.

If $f(x)$ has a local maximum or minimum at $x = c$, then c is a critical number of $f(x)$.

Thus the only points (of the domain) of a function can assume extreme values are critical points and end-points.

Now we are ready to find the absolute maximums or minimums of a continuous function on a closed interval.

The Closed Interval Method

To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical points of f in (a, b).
2. Find the values of f at the end-points of the interval.
3. The largest of these values from steps 1) and 2) is the absolute maximum values; the smallest of these values is the absolute minimum value.

Example

Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1, \quad -\frac{1}{2} \leq x \leq 4.$$

Solution. - Firstly, we find critical points in the open interval $(-\frac{1}{2}, 4)$:

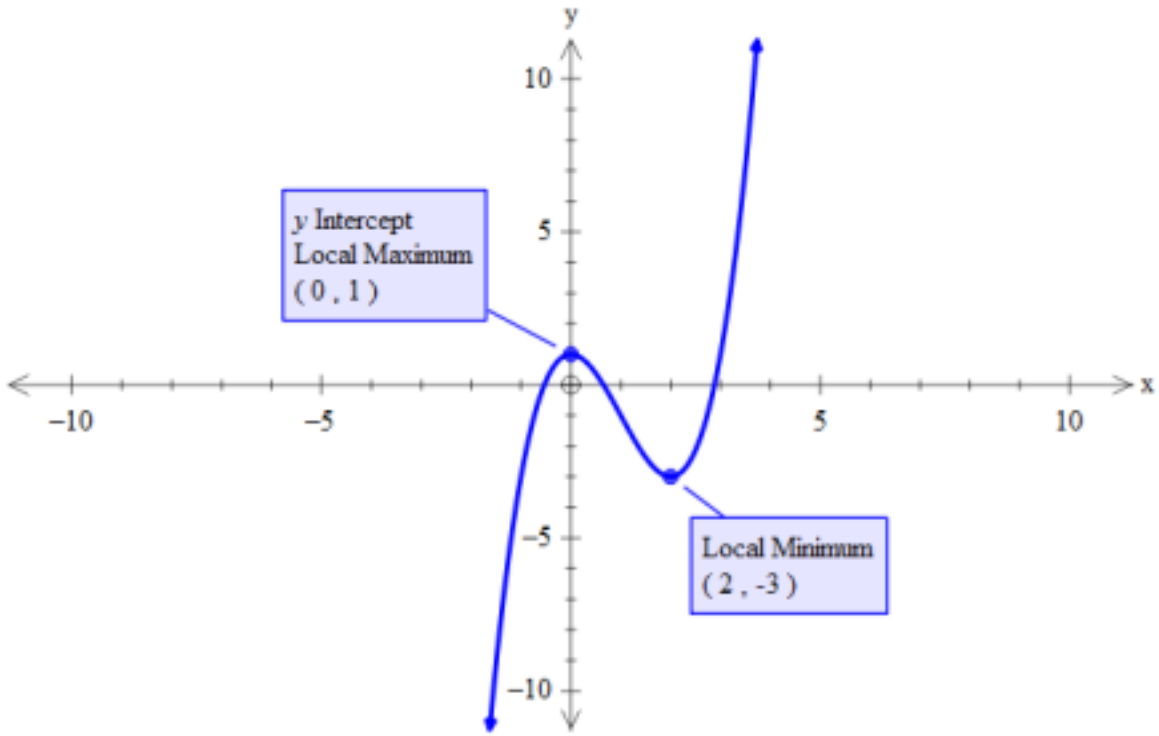
$$f'(x) = 3x^2 - 6x = 3x(x-2) \quad \Longrightarrow \quad x = 0,\ x = 2 : \text{both } \in (-\frac{1}{2}, 4).$$

Hence, $f(0) = 1$ and $f(2) = -3$.

- Next, the values of $f(x)$ at the endpoints are $f\big(-\frac{1}{2}\big) = \frac{1}{8}$ and $f(4) = 17$.

- Lastly, comparing these four values, we see that the absolute maximum value is $f(4) = 17$, and the absolute minimum value is $f(2) = -3$.

The graph of this function looks as follows.



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• Rolle's Theorem

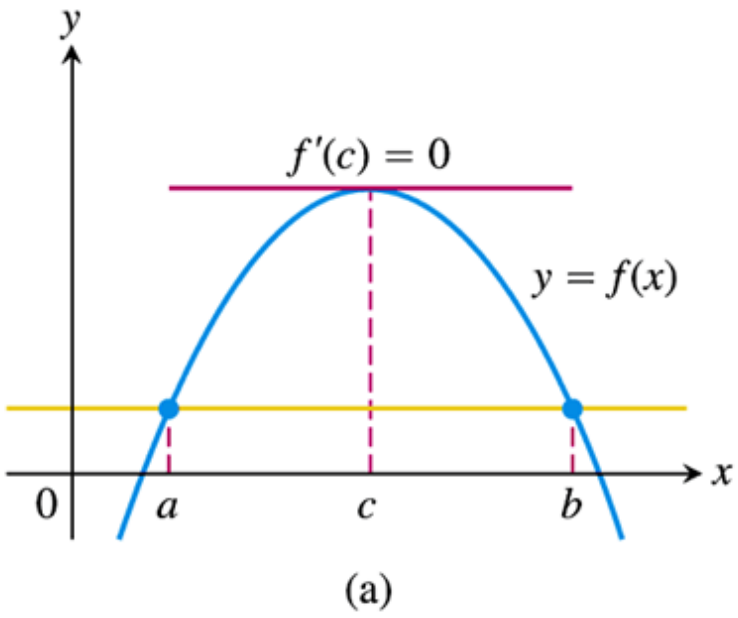
The Mean Value Theorem is essential to Calculus. To arrive at this, we need the following special case known as Rolle's Theorem.

Theorem (Rolle's Theorem)

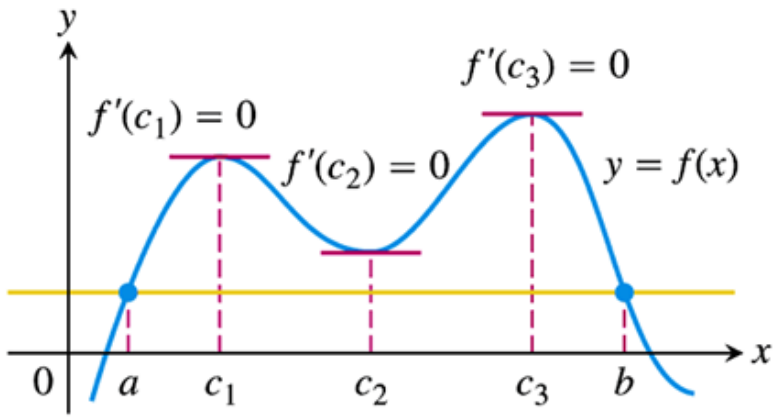
Let f be a function that satisfies the following three conditions:

1. *f is continuous on the closed interval $[a, b]$*
2. *f is differentiable on the open interval (a, b)*
3. *$f(a) = f(b)$.*

Then there is a number $c \in (a, b)$, such that $f'(c) = 0$.



(a)



(b)

Rolle's Theorem says that the graph of a differentiable function has at least one horizontal tangent between any two points where it crosses a horizontal line.

It may have just one in figure a), or it may have more in figure (b).

Example

Prove that the equation

$$x^3 + x - 1 = 0$$

has exactly one real root.

Solution. - First, we use the Intermediate Value Theorem to show that a root exists.

Let $f(x) = x^3 + x - 1$. We have $f(0) = -1 < 0$ and $f(1) = 1 > 0$. Since f is a polynomial, it is continuous, and hence by the Intermediate Value Theorem, there is a number $c \in (0, 1)$ such that $f(c) = 0$. Thus the given equation has a root.

- Next, we use Rolle's Theorem to show that the equation has no other roots, by contradiction.

Assume that there are two roots a and b, i.e. $f(a) = f(b) = 0$, with $a < b$. Since f is a polynomial, it is differentiable on (a, b) and continuous on $[a, b]$. Hence, by Rolle's Theorem, there is a number $c \in (a, b)$ such that $f'(c) = 0$.

But $f'(x) = 3x^2 + 1 \geq 1$ for all x, which shows that $f'(x)$ can never be 0. This gives a contradiction.

Therefore, the equation can't have more than one real root.

Combining the two facts yields that the given equation has exactly one real root.

• Mean Value Theorem

Using Rolles' Theorem we can prove the following important result.

Theorem (The Mean Value Theorem (Lagrange's Theorem))

Let f be a function that satisfies the following conditions:

1. *f is continuous on the closed interval $[a, b]$*
2. *f is differentiable on the open interval (a, b).*

Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a},$$

or equivalently,

$$f(b) - f(a) = f'(c)(b - a).$$

We can interpret its geometrical meaning.

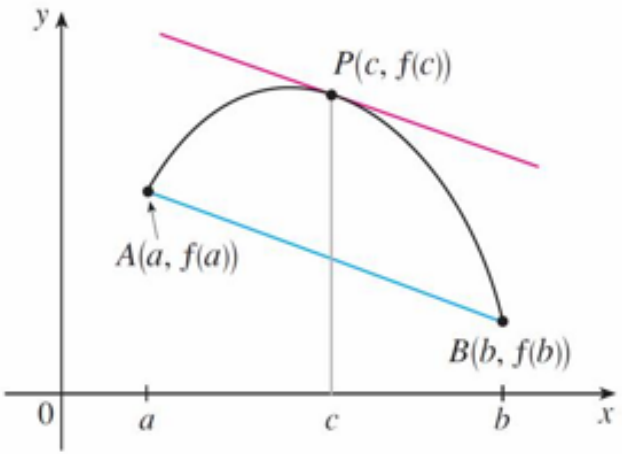


Figure 3

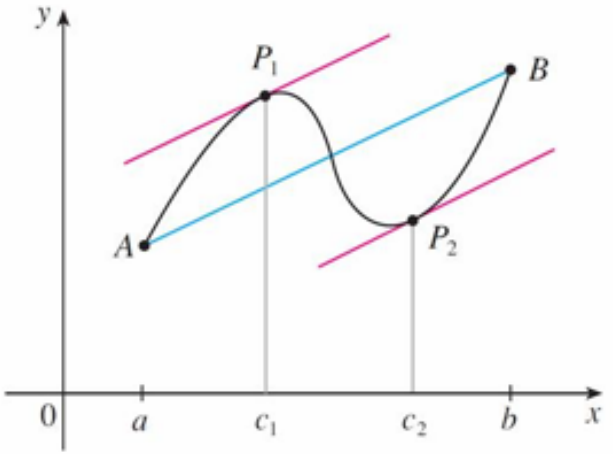


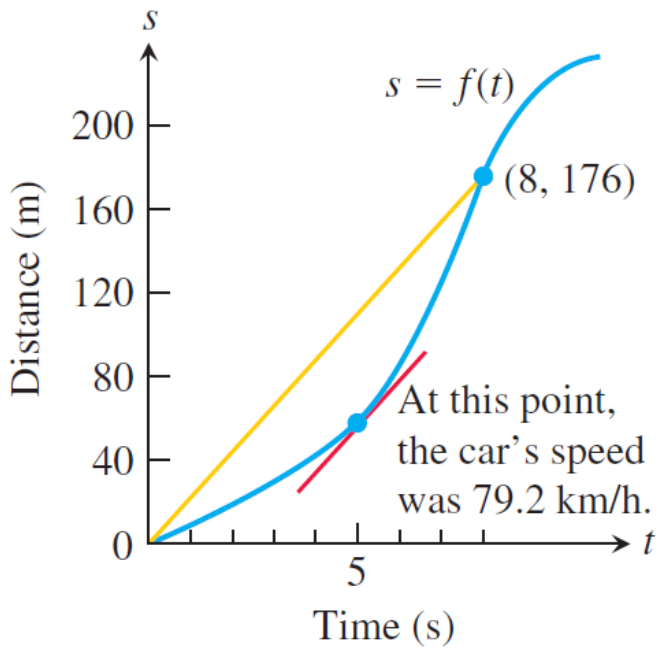
Figure 4

The slope of the secant line AB is $m_{AB} = \dfrac{f(b) - f(a)}{b - a}$.

The Mean Value Theorem indicates there exists (at least one) point $P(c, f(c))$ on the graph, such that the tangent to the curve at P is parallel to the secant AB.

Exercise (A physical interpretation)

A car accelerating from 0 takes 8 sec to go 176 m, and hence its average velocity for the 8-sec interval is $\frac{176}{8} = 22$ m/s. Using the Mean Value Theorem to show that at some interior point during the acceleration, the speedometer must read exactly 79.2 km/h (22 m/s).



As consequences of the Mean Value Theorem, we have the following results.

Theorem

If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b).

Theorem

If $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f - g$ is constant on (a, b); that is, $f(x) = g(x) + C$, where C is constant.

Note. Care must be taken in applying the theorem above. For example,

$$f(x) = \frac{x}{|x|} = \begin{cases} 1 & x > 0 \\ -1 & x < 0. \end{cases}$$

The domain of f is $D_f = \{x : x \neq 0\}$ and $f'(x) = 0$ for all $x \in D$. But f is obviously not a constant function.

Does it contradict the theorem?

The answer is NO, because D is not an interval!

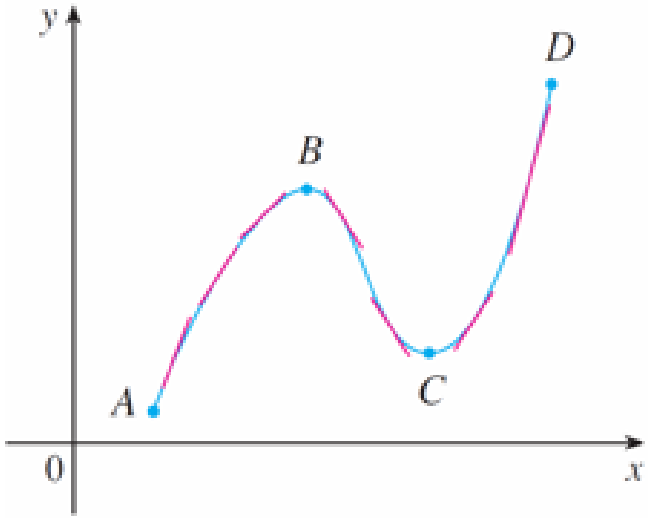
Notice that f is constant on the interval $(0, \infty)$ and also on the interval $(-\infty, 0)$.

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• What does f' say about f?

Derivative of $f(x)$ can tell us where a function is increasing/decreasing!



Between A and B, C and D, tangent lines have positive slopes, i.e. $f'(x) > 0$; also $f(x)$ is increasing, while between B and C, tangent lines have negative slope, i.e. $f'(x) < 0$; also $f(x)$ is decreasing.

We can use the Mean Value Theorem to prove the following result.

Increasing-Decreasing (or Monotonic) Test

- If $f'(x) > 0$ on an interval, then $f(x)$ is increasing on that interval
- If $f'(x) < 0$ on an interval, then $f(x)$ is decreasing on that interval

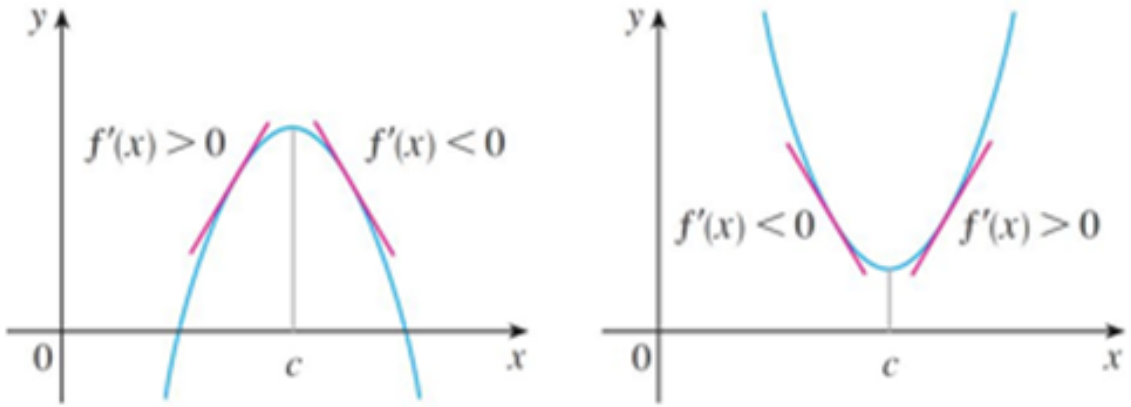
Now we have the following test.

The First Derivative Test

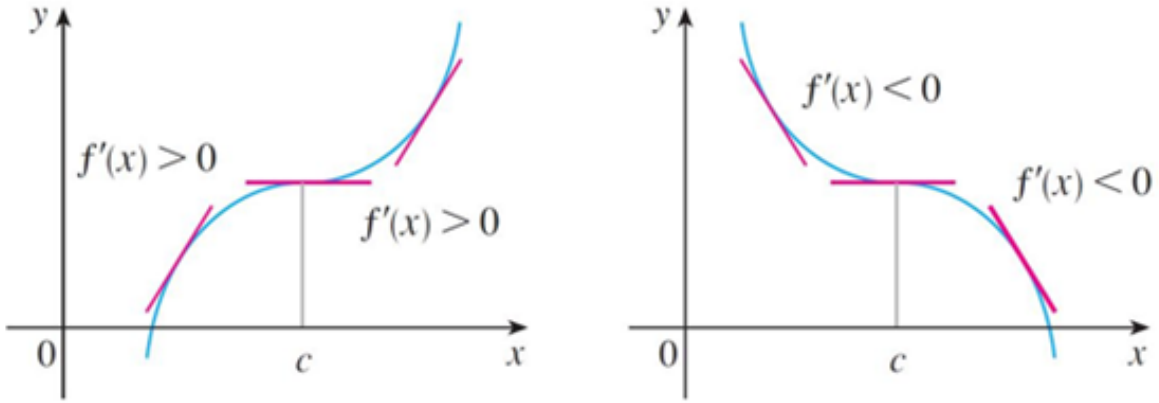
Suppose c is a critical point of a continuous function f.

- If f' changes from $+$ to $-$ at c, then f has a local max at c.
- If f' changes from $-$ to $+$ at c, then f has a local min at c.
- If f' is $+$ to the right and left of c, or $-$ to the left and right of c, then f has no local max or minimum at c.

- Local maximum and local minimum:



- No maximum or minimum:



Example

Find the local maximum and minimum values of the function

$$g(x) = x + 2\sin x, \quad 0 \leq x \leq 2\pi.$$

Solution. We start by finding the critical points. The derivative of g is $g'(x) = 1 + 2\cos x$, and so

$$g'(x) = 0 \quad \Longleftrightarrow \quad \cos x = -\frac{1}{2} \quad \Longleftrightarrow \quad x = \frac{2\pi}{3}, x = \frac{4\pi}{3}.$$

Since g is differentiable everywhere, these are only critical points.

We split the domain into intervals by the critical points. Within each interval, $g'(x)$ is either always positive or always negative and so we analyze g in the following table.

Interval	$g'(x) = 1 + 2\cos x$	g
$0 < x < 2\pi/3$	$+$	increasing on $(0, 2\pi/3)$
$2\pi/3 < x < 4\pi/3$	$-$	decreasing on $(2\pi/3, 4\pi/3)$
$4\pi/3 < x < 2\pi$	$+$	increasing on $(4\pi/3, 2\pi)$

According to the table, by the First Derivative Test, we obtain

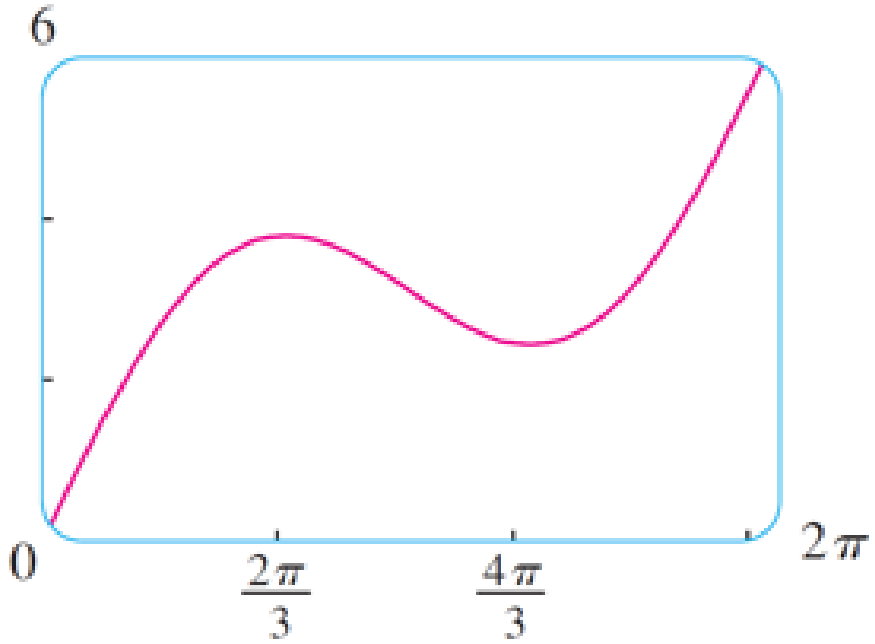
- $g(x)$ has a local maximum at $\dfrac{2\pi}{3}$ and the local maximum value is

$$g(2\pi/3) = \frac{2\pi}{3} + 2\sin\frac{2\pi}{3} \approx 3.83,$$

- $g(x)$ has a local minimum at $\dfrac{4\pi}{3}$, and the local minimum value is

$$g(4\pi/3) = \frac{4\pi}{3} + 2\sin\frac{4\pi}{3} \approx 2.46.$$

The graph of $g(x)$ below supports our conclusion.

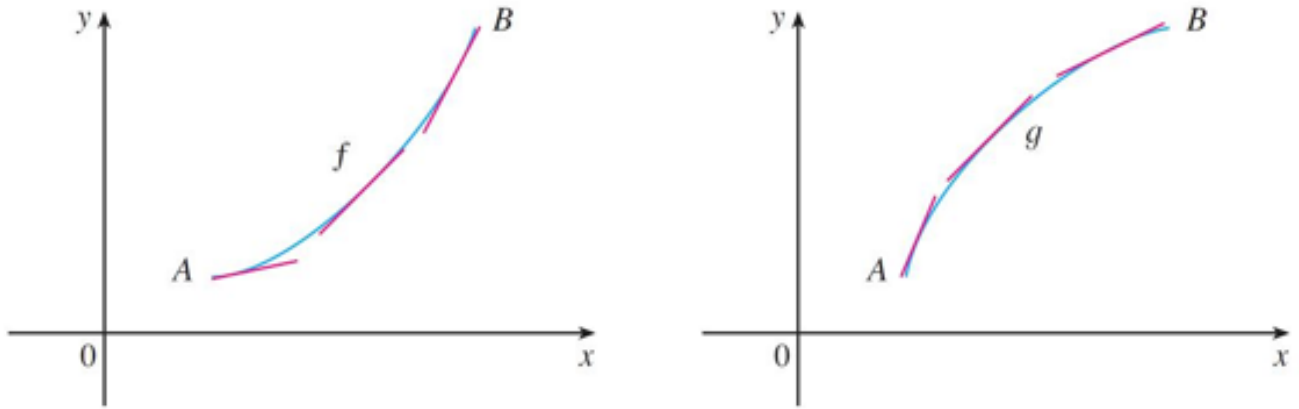


$g(x) = x + 2 \sin x$

In the Figure below, graphs of two increasing functions on (a, b). Both graphs join point A to point B but they look different because they bend in different directions.

We check the tangents to these curves and see that

- Left figure: the curve lies above the tangents
- Right figure: the curve lies below the tangents.

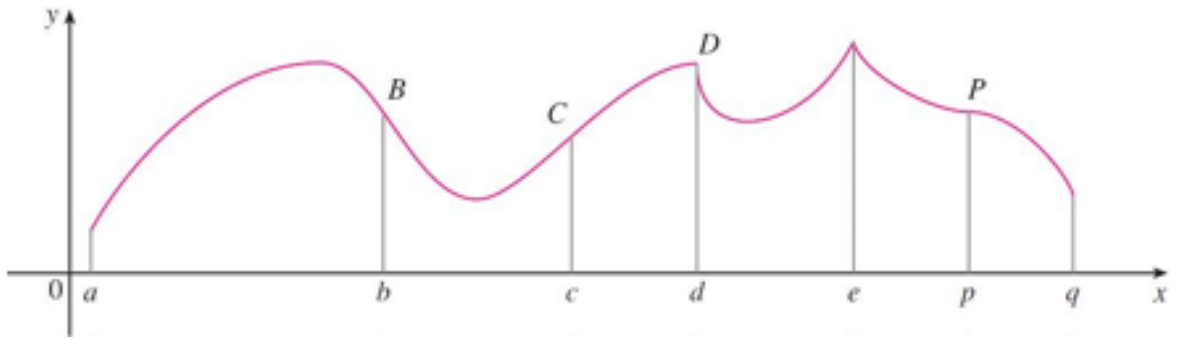


• What does f'' say about f?

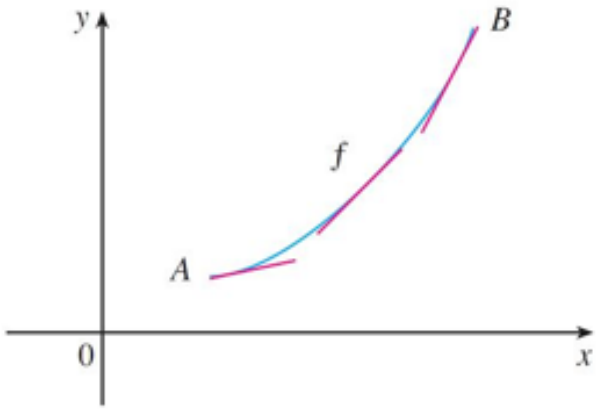
Definition.

- If the graph of f lies above all of its tangents on an interval I, then it is called concave up (or upward) in I
- If the graph of f lies below all of its tangents on an interval I, then it is called concave down (or downward) in I.

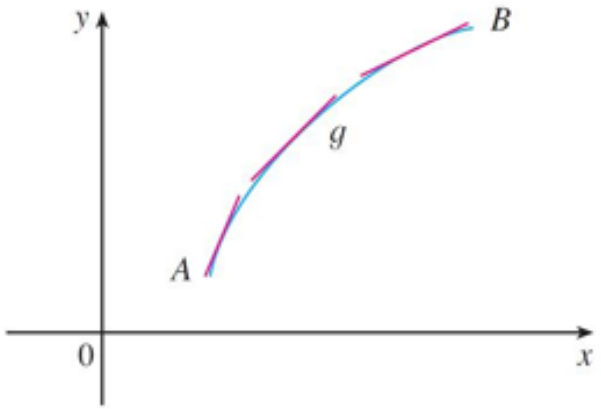
Figure below shows the graph of a function that is concave up on the intervals (b, c), (d, e), and (e, p), and is concave down on the intervals (a, b), (c, d), and (p, q).



Look at this figure.



Concave upward



Concave downward

We can notice the following

- Left figure: the slope of the tangent increases, meaning that f' is an increasing function and therefore its derivative f'' is positive.

- Right figure: the slope of the tangent decreases, so f' decreases and therefore f'' is negative.

So we have the following result.

Concavity Test

- If $f''(x) > 0$ for all x in I, then graph of f is concave up on I.
- If $f''(x) < 0$ for all x in I, then graph of f is concave down on I

It can happen that a graph of a function changes its concavity passing through a point. We give a special name to this point.

• Second Derivative Test

Definition. A point P on a curve $y = f(x)$ is called an inflection point if f is continuous at this point, and the curve changes from concave up to concave down or vice-versa.

We have the following result.

The Second Derivative Test

Suppose f'' are continuous near c.

- If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c
- If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

Note.

- The Second Derivative Test is inconclusive when $f''(c) = 0$.

  In other words, at such a point there might be a maximum, or a minimum, or neither.

- This test also fails when $f''(c)$ does not exist.

  In such cases the First Derivative Test must be used.

Note. In fact, even when both tests apply, the First Derivative Test is often the easier one to use.

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• Asymptotes

For curve sketching of $y = f(x)$, we need notion of asymptotes.

If the distance between the graph of a function and some fixed line approaches zero as a point on the graph moves increasingly far from the origin, we say that

- the graph approaches the line asymptotically, and
- the line is an asymptote of the graph.

- Vertical asymptotes

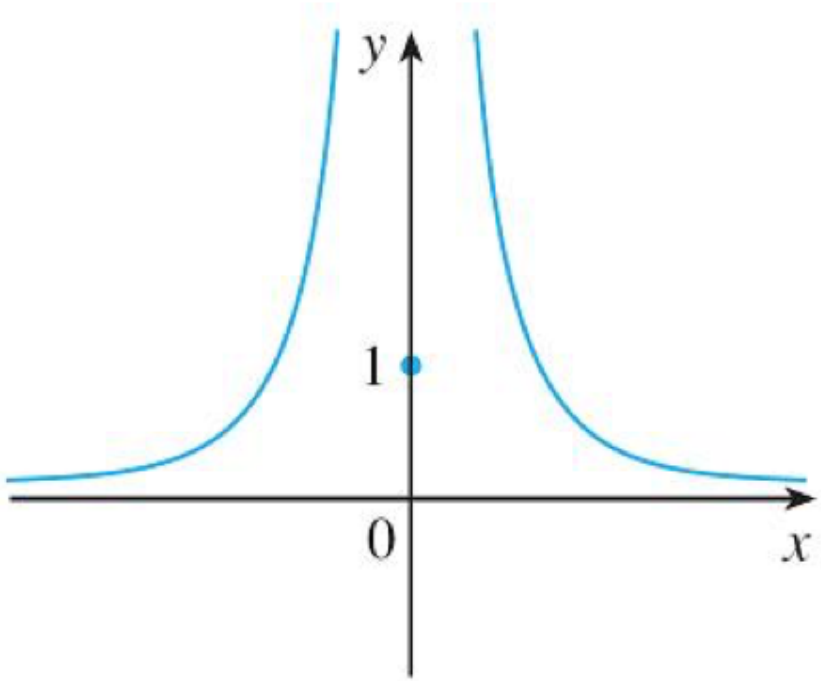
Definition. The vertical line $x = a$ is called a vertical asymptote of the graph of $y = f(x)$ if either of the following is true

$$\lim_{x\to a} f(x) = \infty, \qquad \lim_{x\to a^-} f(x) = \infty, \qquad \lim_{x\to a^+} f(x) = \infty,$$

$$\lim_{x\to a} f(x) = -\infty, \qquad \lim_{x\to a^-} f(x) = -\infty, \qquad \lim_{x\to a^+} f(x) = -\infty.$$

For instance, the line $x = 0$ is a vertical asymptote of the function

$$y = f(x) = \begin{cases} \dfrac{1}{x^2}, & x \neq 0 \\ 1, & x = 0. \end{cases}$$



- Horizontal asymptotes

Definition. The horizontal line $y = b$ is called a horizontal asymptote of the graph of a function $y = f(x)$ if either of the following is true

$$\lim_{x \to \infty} f(x) = b, \qquad \lim_{x \to -\infty} f(x) = b.$$

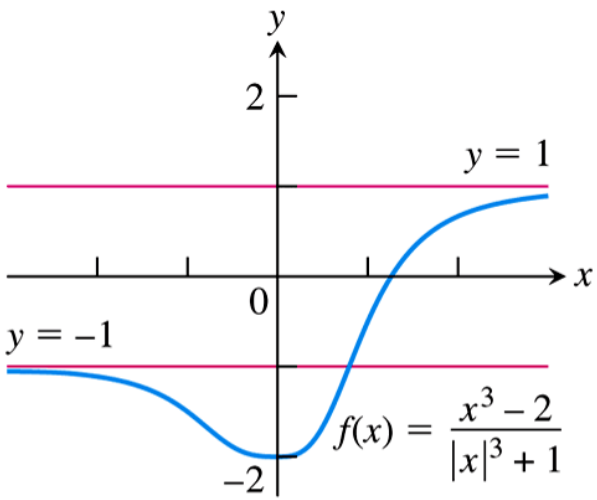
Example

Find the horizontal asymptotes of the graph of $f(x) = \dfrac{x^3 - 2}{|x|^3 + 1}$

Solution. - For $x \geq 0$: $\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \dfrac{x^3 - 2}{x^3 + 1} = \lim\limits_{x\to\infty} \dfrac{1 - \frac{2}{x^3}}{1 + \frac{1}{x^3}} = 1.$

- For $x < 0$: $\lim\limits_{x\to-\infty} f(x) = \lim\limits_{x\to\infty} \dfrac{x^3 - 2}{-x^3 + 1} = \lim\limits_{x\to\infty} \dfrac{1 - \frac{2}{x^3}}{-1 + \frac{1}{x^3}} = -1.$

So the horizontal asymptotes are $y = \pm 1$. Notice that the graph crosses the horizontal asymptote $y = -1$ for a positive value of x.



- Oblique asymptotes

In the case when $f(x)$ is a rational function

$$f(x) = \frac{P(x)}{Q(x)}, \quad \text{where } P \text{ and } Q \text{ are polynomials,}$$

we have a notion of *oblique asymptotes*.

If $\deg(P) = \deg(Q) + 1$, then the graph has an oblique asymptote.

We can find an equation for the asymptote as follows:

Dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \to \pm\infty$. That is,

$$f(x) = \underbrace{(mx + b)}_{\text{an oblique asymptote}} + R(x), \quad \text{with } R(x) \to 0 \text{ as } x \to \pm\infty.$$

Example

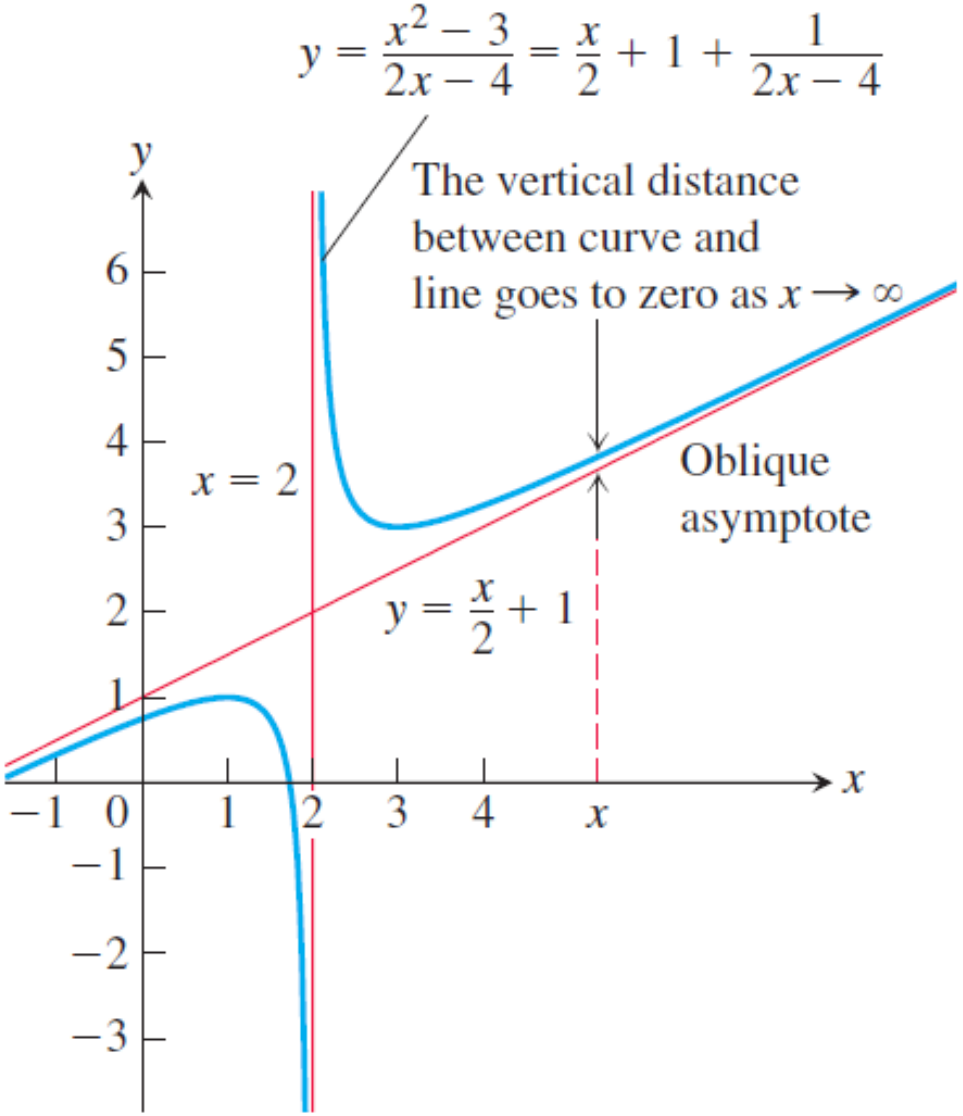
Find the oblique asymptote of the graph of $f(x) = \dfrac{x^2 - 3}{2x - 4}$.

Solution. Dividing $2x - 4$ into $x^2 - 3$, we get

$$f(x) = \frac{x^2 - 3}{2x - 4} = \left(\frac{x}{2} + 1\right) + \left(\frac{1}{2x - 4}\right).$$

Since the remainder $\dfrac{1}{2x - 4} \to 0$ as $x \to \pm\infty$, the linear function (a line) $y = \dfrac{x}{2} + 1$ is an oblique asymptote of the graph of f, both to the right and to the left.

The graph of the function f looks as follows.



• Guidelines for a Curve Sketching

These guidelines provide all the information you need to make a sketch of a function, that captures its most important properties. You should note that not every item is relevant.

A. Domain

It is often useful to start by determining the domain D of $f(x)$, on which $f(x)$ is defined.

B. Intercepts

- The y-intercept is $f(0)$.
- To find the x-intercepts, we set $y = 0$ and solve for x. (Omit this step, if the function is difficult to solve.)

C. Symmetry

- Even function if $f(-x) = f(x)$, for all $x \in D$.
- Odd function if $f(-x) = -f(x)$, for all $x \in D$.
- Periodic function if $f(x + p) = f(x)$, for all $x \in D$.

D. Asymptotes

- Horizontal Asymptotes. If either $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, then the line $y = L$ is a horizontal asymptote.

- Vertical Asymptotes. If at least one of the following statements is true

$$\lim_{x\to a^+} f(x) = \infty, \quad \lim_{x\to a^-} f(x) = \infty, \quad \lim_{x\to a^+} f(x) = -\infty, \quad \lim_{x\to a^-} f(x) = -\infty,$$

then the line $x = a$ is a vertical asymptote.

If $f(a)$ is not defined, but a is an endpoint of the domain of $f(x)$, then you should check whether or not $\lim_{x\to a^-} f(x)$ or $\lim_{x\to a^+} f(x)$, is infinite.

- Oblique Asymptotes. In case $f(x)$ is a rational function, if

$$\lim_{x\to\infty}[f(x)-(mx+b)]=0,$$

then the line $y=mx+b$ ($m\neq 0$) is an oblique asymptote.

E. Intervals of Increase or Decrease

Compute $f'(x)$ and find

- the intervals on which $f'(x)>0$ (f is increasing)
- the intervals on which $f'(x)<0$ (f is decreasing).

Here the *Increasing-Decreasing Test* is used.

F. Local Maximum and Minimum Values

- Find the critical points of $f(x)$ (those points c where $f'(c) = 0$ or $f'(c)$ does not exist).

- Then use the *First Derivative Test*. If f' changes at c from $+$ to $-$, then $f(c)$ is a local maximum. If f' changes at c from $-$ to $+$, then $f(c)$ is a local minimum.

Although it is usually preferable to use the First Derivative Test, you can use the *Second Derivative Test*. If $f'(c) = 0$ and $f''(c) \neq 0$, then $f''(c) > 0$ implies that $f(c)$ is a local minimum, whereas $f''(c) < 0$ implies that $f(c)$ is a local maximum.

G. Concavity and Points of Inflection

Compute $f''(x)$ and use the *Concavity Test*. The curve is concave upward where $f''(x) > 0$, and concave downward where $f''(x) < 0$. Inflection points occur where the direction of concavity changes.

H. Sketch the Curve

- Using the information in items **A - G** to draw the graph.
- Sketch the asymptotes in **D** as dashed lines.
- Plot the intercepts, maximum/minimum, and inflection points.
- Make the curve pass through these points, rising and falling according to **E**, with concavity according to **G**, and approaching the asymptotes in **D**.
- If additional accuracy is required near any point, you can compute the value of its derivative. The tangent indicates the direction in which the curve proceeds.

Exercise

Sketch the graph of $f(x) = \dfrac{x^3}{x^2+1}$.

Answer.

